

Case Metadata

Field	Value
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Case ID:	CS-2025-001
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Case Type:	Network Service Discovery
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Reported by:	Splunk SIEM
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Analyst:	Ali YILMAZ
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Date:	2025-09-30
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Severity:	High
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Status:	Closed
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Environment:	Production
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Detection Rule Details

Rule Name: Network Service Discovery

Rule Description: An adversary uses a combination of techniques to determine the state of the ports on a remote target. Any service or application available for TCP or UDP networking will have a port open for communications over the network. Although common services have assigned port numbers, services and applications can run on arbitrary ports. Additionally, port scanning is complicated by the potential for any machine to have up to 65535 possible UDP or TCP services. The goal of port scanning is often broader than identifying open ports, but also give the adversary information concerning the firewall configuration. Depending upon the method of scanning that is used, the process can be stealthy or more obtrusive, the latter being more easily detectable due to the volume of packets involved, anomalous packet traits, or system logging. Typical port scanning activity involves sending probes to a range of ports and observing the responses. There are four port statuses that this type of attack aims to identify: open, closed, filtered, and unfiltered. For strategic purposes it is useful for an adversary to distinguish between an open port that is protected by a filter vs. a closed port that is not protected by a filter. Making these fine grained distinctions is requires certain scan types. Collecting this type of information tells the adversary which ports can be attacked directly, which must be attacked with filter evasion techniques like fragmentation, source port scans, and which ports are unprotected (i.e. not firewalled) but aren't hosting a network service. An adversary often combines various techniques in order to gain a more complete picture of the firewall filtering mechanisms in place for a host.

Correlation Rule (SPL):

```
| tstats dc(All_Traffic.dest_port) as scannedPortCount  
from datamodel=Network_Traffic.All_Traffic  
by _time span=10m, All_Traffic.src_ip  
| rename All_Traffic.* as *  
| where scannedPortCount > 50
```

Triggered Field(s):

srcip, dstip, dstport, _time

(Alarm was triggered because the same srcip attempted connections to more than 50 distinct destination ports within a 10-minute window.)

Detection Summary

An adversary may attempt to identify running services on remote systems by scanning multiple ports across one or more hosts. This behavior often precedes lateral movement or exploitation. The detected activity matches known scanning patterns, where a single source IP rapidly probed multiple destination hosts and ports within a short time window.

Analyst Investigation

- The scanning pattern (multiple dstip and dstport within short time) strongly indicates port scanning activity.
- The source IP belongs to an external network. Preliminary OSINT check VirusTotal shows suspicious reputation.
- No scheduled vulnerability scan or internal asset ownership identified for this IP.
- Behavior aligns with MITRE ATT&CK T1046 (Network Service Discovery).

Decision:  **True Positive** (unauthorized port scanning attempt).

Investigation Steps:

- Queried Splunk for similar events within the last 24 hours.
- Checked IOC reputation (domain/IP) in VirusTotal

MITRE ATT&CK Mapping

- T1046 – Network Service Discovery

 **True Positive** (unauthorized port scanning attempt).

Recommended Actions (SOC Level)

Block 109.205.211.129 at perimeter firewall.

Review firewall logs for additional attempts from the same ASN.

Conduct threat hunting for related activity in EDR/SIEM (look for lateral movement attempts on the scanned hosts).

Update detection logic to enrich alerts with ASN/WHOIS reputation.

Organizational / HR Actions

- Notify network team and service owners of affected hosts (96.73.98.x subnet).
- Document incident in the central tracker and link to whitelist/exceptions list if legitimate scanner is identified later.

Business Impact

Unauthorized reconnaissance against production assets increases the risk of exploitation of exposed services (RDP, SMB, SQL, etc.).

If scanning is successful, adversary may discover vulnerable services, leading to lateral movement or data exfiltration.

Evidence in Case

Splunk search query results and screenshot (showing srcip, dstip, dstport, and event counts)

 Firewall / Network traffic logs indicating multiple connection attempts from source 109.205.211.129

DNS query logs (to validate whether scanning activity also attempted name resolution)

Proxy traffic logs (to confirm if outbound traffic patterns deviate from baseline)

IOC reputation check results (VirusTotal, AbuseIPDB, Talos lookups for 109.205.211.129)

MITRE ATT&CK Navigator mapping highlighting technique **T1046 – Network Service Discovery**

Incident Review

Saved Filters: [Select...] Tag: [Add tags...] Urgency: [Select...] Status: [Select...] Owner: [Select...] Security Domain: [Select...] Type: [Select...] Search Time or Associations: Correlation Score: [Select...] Time: [Time] Last 15 min... [v]

[Save new filters] [Update] [Clear all] [Submit] Time Range: [Last 15 minutes]

1 Notables [Unselect all] | [Edit Selected] | [Edit All Matching Events (1)] | Add Selected to Investigation [20 per page v] [Refresh Q]

☐	i	Title	Risk Object	Risk Score	Risk Events	Type	Time	Disposition	Security Domain	Urgency	Status	Owner	Actions
☒		Network Service Discovery		--	--	Notable	Today, 4:00 AM	Undetermined	Network	▲ Medium	New	unassigned	v

Description:

An adversary uses a combination of techniques to determine the state of the ports on a remote target. Any service or application available for TCP or UDP networking will have a port open for communications over the network. Although common services have assigned port numbers, services and applications can run on arbitrary ports. Additionally, port scanning is complicated by the potential for any machine to have up to 65535 possible UDP or TCP services. The goal of port scanning is often broader than identifying open ports, but also give the adversary information concerning the firewall configuration. Depending upon the method of scanning that is used, the process can be stealthy or more obtrusive, the latter being more easily detectable due to the volume of packets involved, anomalous packet trails, or system logging. Typical port scanning activity involves sending probes to a range of ports and observing the responses. There are four port statuses that this type of attack aims to identify: open, closed, filtered, and unfiltered. For strategic purposes it is useful for an adversary to distinguish between an open port that is protected by a filter vs. a closed port that is not protected by a filter. Making these fine grained distinctions requires certain scan types. Collecting this type of information tells the adversary which ports can be attacked directly, which must be attacked with filter evasion techniques like fragmentation, source port scans, and which ports are unprotected (i.e. not firewalled) but aren't hosting a network service. An adversary often combines various techniques in order to gain a more complete picture of the firewall filtering mechanisms in place for a host.

Additional Fields

Annotations	Value	Action
Annotation Framework	mstrle_attack	v
Severity	high	v
Source IP Address	109.205.211.129	v

Event Details:

event_id	FC5EFEEF-DE6E-4C5A-902C-4CB2803883E5@notable@65d42e04ae42e531da090c6916af2fb
event_hash	65d42e04ae42e531da090c6916af2fb
eventtype	modnotable_results
Short ID	Create Short ID

Related Investigations:

Currently not investigated.

Correlation Search:

Network - Network Service Discovery - Rule [L](#)

History:

View all review activity for this Notable Event [L](#)

Contributing Events:

Find Network Logs [L](#)

Response	Mode	Time	User	Status
Risk Analysis	saved	2025-09-30T04:00:23-0400	admin	✓ success
Notable	saved	2025-09-30T04:00:22-0400	admin	✓ success

View Adaptive Response Invocations [L](#)

Next Steps:

No next steps defined.

splunk>enterprise Apps ▾ All Yilmaz ▾ 27 Messages ▾ Settings ▾ Activity ▾ Help ▾ Q Find

Security Posture ▾ Incident Review Investigations Security Intelligence ▾ Security Domains ▾ Cloud Security ▾ Audit ▾ Search ▾ Configure ▾ Enterprise Security

New Search

Save As ▾ Create Table View Close

index=fortinet srcip=109.205.211.129 | table srcip dstip dstport _ time | dedup dstport

Date time range ▾ 🔍

✓ 112 events (9/30/25 3:40:00.000 AM to 9/30/25 3:50:00.000 AM) No Event Sampling ▾ Job ▾ || ▮ ↻ ⚙️ 📄 Smart Mode ▾

Events Patterns **Statistics (62)** Visualization

20 Per Page ▾ Format Preview ▾ < Prev 1 2 3 4 Next >

srcip ▴	dstip ▴	dstport ▴	_time ▴
109.205.211.129	96.73.98.189	12164	03:43:09
109.205.211.129	96.73.98.186	12382	03:43:09
109.205.211.129	96.73.98.188	39792	03:43:02
109.205.211.129	96.73.98.187	19079	03:42:59
109.205.211.129	96.73.98.186	12900	03:42:50
109.205.211.129	96.73.98.188	28789	03:42:49
109.205.211.129	96.73.98.184	13432	03:42:47
109.205.211.129	96.73.98.187	24754	03:42:27
109.205.211.129	96.73.98.184	24900	03:42:22
109.205.211.129	96.73.98.185	29384	03:42:21
109.205.211.129	96.73.98.184	19136	03:42:08
109.205.211.129	96.73.98.184	10883	03:42:04
109.205.211.129	96.73.98.188	12326	03:41:34
109.205.211.129	96.73.98.189	26217	03:41:29
109.205.211.129	96.73.98.188	34359	03:41:19
109.205.211.129	96.73.98.185	17921	03:41:11
109.205.211.129	96.73.98.185	27164	03:40:37
109.205.211.129	96.73.98.185	32862	03:40:33
109.205.211.129	96.73.98.184	12583	03:40:27
109.205.211.129	96.73.98.186	25756	03:40:24

splunk-enterpriseApps

All Yilmaz999MessagesSettingsActivityHelpFind

Security PostureIncident ReviewInvestigationsSecurity IntelligenceSecurity DomainsCloud SecurityAuditSearchConfigureEnterprise Security

New Search

Save AsCreate Table ViewClose

Index=fort.list, srcip=189.285.211.129

Date time range

112 events (9/30/25 3:40:00.000 AM to 9/30/25 3:50:00.000 AM)No Event SamplingJob

Events (112)PatternsStatisticsVisualization

Format TimelineZoom OutZoom to SelectionDeselect1 minute per column

ListFormat20 Per Page

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

a action 1

a app 30

a bytes 1

a bytes_in 1

a bytes_out 1

a craction 1

a crlevel 1

a crscore 1

a date 1

a date_hour 1

a date_minute 1

a date_minute 10

a date_month 1

a date_second 53

a date_weekday 1

a date_year 1

a date_zone 1

a dest 6

a dest_interface 1

a dest_ip 6

a dest_port 62

a dest_zone 1

a devid 1

a devname 1

a dscountry 1

a dstintf 1

a dstintfrole 1

a dstip 6

a dstport 62

a duration 1

a dvc 1

a eventtime 100+

a eventtype 5

a fnt_action 1

a fnt_index 1

a level 1

a linecount 1

a logid 1

a msg 1

a packets 1

a packets_in 1

a packets_out 1

a policyid 1

a policytype 1

a product 1

a product_version 1

a proto 1

a protocol 1

a protocol_version 1

a rcvbyte 1

a rcvdpkt 1

a rule_id 1

a sentbyte 1

a sentpkt 1

a service 30

a session_id 100+

a sessionid 100+

a splunk_server 1

a src 1

a src_interface 1

a src_ip 1

a src_port 1

a src_translated_port 1

a src_zone 1

a sccountry 1

a scrcntf 1

a scrcntfrole 1

a srcip 1

a srcport 1

a subtype 1

a tag 3

a tag:eventtype 3

a time 100+

a timeendpos 1

a timestamppos 1

a transdp 1

a transport 1

a type 1

a tz 1

a vd 1

a vendor 1

a vendor_action 1

a vendor_product 1

+ Extract New Fields

Time

9/30/25

3:49:58.000 AM

Event

date=2025-09-30 time=03:49:58 devname="CNL" devid="FGT60FTK19020806" eventtime=1759218597996089360 tz="-0400" logid="0001000014" type="traffic" subtype="local" level="notice" vdr="root" srcip=189.285.211.129 srcport=50605 scrcntf="wan" scrcntfrole="wan" dstip=96.73.98.188 dstport=11658 dstintf="root" ds

data=log/splunk/syslog/1011254/splunk-collector-2025-09-30-03.log

srcintfrole="undefined" sccountry="Poland" dscountry="United States" sessionid=5199511 protocol="tcp" action="deny" policytype="local-in-policy" serv

icer="00MA" transdp="noop" app="00MA" duration=0 sentbyte=0 rcvbyte=0 sentpkt=0 rcvdpkt=0 craction=262144 crlevel="low" msg="Connection Failed"

4

EVENT ACTIONS

Type	Field	Value	Actions
Selected	host	1011.254	
	source	/data/log/splunk/syslog/1011254/splunk-collector-2025-09-30-03.log	
	sourcetype	fortigate_traffic	
Event	action	blocked	
	app	OOMA	
	bytes	0	
	bytes_in	0	
	bytes_out	0	
	craction	262144	
	crlevel	low	
	crcore	5	
	date	2025-09-30	
	dest	96.73.98.188	
	dest_interface	root	
	dest_ip	96.73.98.188	
	dest_port	11658	
	dest_zone	undefined	
	devid	FGT60FTK19020806	
	devname	CNL	
	dscountry	United States	
	dstintf	root	
	dstintfrole	undefined	
	dstip	96.73.98.188	
	dstport	11658	
	duration	0	
	dvc	CNL	
	eventtime	1759218597996089360	
	eventtype	errOr (error)	
	fnt_action	fnt_fortigate	
	fnt_index	fnt_fortigate_traffic (communicate network)	
	fnt_level	notice	
	fnt_logid	0001000014	
	fnt_msg	Connection Failed	
	fnt_packets	0	
	fnt_packets_in	0	
	fnt_packets_out	0	
	fnt_policyid	0	
	fnt_policytype	local-in-policy	
	fnt_product	Firewall	
	fnt_product_version	50	
	fnt_proto	6	
	fnt_protocol	ip	
	fnt_protocol_version	ipV4	
	fnt_rcvbyte	0	
	fnt_rcvdpkt	0	
	fnt_rule_id	0	
	fnt_sentbyte	0	
	fnt_sentpkt	0	
	fnt_service	OOMA	
	fnt_session_id	5199511	
	fnt_sessionid	5199511	
	fnt_src	109.205.211.129	
	fnt_src_interface	wan1	
	fnt_src_ip	109.205.211.129	
	fnt_src_port	50605	
	fnt_src_translated_port	50605	
	fnt_src_zone	wan	
	fnt_sccountry	Poland	
	fnt_scrcntf	wan1	
	fnt_scrcntfrole	wan	
	fnt_srcip	109.205.211.129	
	fnt_srcport	50605	
	fnt_subtype	local	
	fnt_tag	communicate	
	fnt_time	03:49:58	
	fnt_transdp	noop	
	fnt_transport	tcp	
	fnt_type	traffic	
	fnt_tz	-0400	
	fnt_vd	root	
	fnt_vendor	Fortinet	
	fnt_vendor_action	deny	
	fnt_vendor_product	Fortinet Firewall	
Time	_time	2025-09-30T03:49:58.000-04:00	
Default	index	fortinet	
	linecount	1	
	splunk_server	splunk-prod	

109.205.211.129

5/95

Community Score

5/95 security vendors flagged this IP address as malicious

Reanalyze Similar More

109.205.211.129 (109.205.211.0/24)

AZ

Last Analysis Date

AS 201814 (MEVSPACE sp. z o.o.)

14 days ago

DETECTION

DETAILS

RELATIONS

COMMUNITY

Security vendors' analysis

Do you want to automate checks?

ArcSight Threat Intelligence	Malware	CINS Army	Malicious
Fortinet	Malware	MalwareURL	Malware
SOCRadar	Malware	alphaMountain.ai	Suspicious
AlphaSOC	Suspicious	CyRadar	Suspicious
Gridinsoft	Suspicious	Abusix	Clean
Acronis	Clean	ADMINUSLabs	Clean

MITRE | ATT&CK

Matrices Tactics Techniques Defenses CTI Resources Benefactors Blog Search

ATT&CKcon 6.0 is coming October 14-15 in McLean, VA and live online. Tickets are available now!

TECHNIQUES

Network Service Discovery

Network Share Discovery

Network Sniffing

Password Policy Discovery

Peripheral Device Discovery

Permission Groups Discovery

Process Discovery

Query Registry

Remote System Discovery

Home > Techniques > Enterprise > Network Service Discovery

Network Service Discovery

Adversaries may attempt to get a listing of services running on remote hosts and local network infrastructure devices, including those that may be vulnerable to remote software exploitation. Common methods to acquire this information include port, vulnerability, and/or wordlist scans using tools that are brought onto a system.^[1]

Within cloud environments, adversaries may attempt to discover services running on other cloud hosts. Additionally, if the cloud environment is connected to a on-premises environment, adversaries may be able to identify services running on non-cloud systems as well.

Within macOS environments, adversaries may use the native Bonjour application to discover services running on other macOS hosts within a network. The Bonjour mDNSResponder daemon automatically registers and advertises a host's registered services on the network. For example, adversaries can use a mDNS query (such as `dns-sd -B _ssh._tcp .`) to find other systems broadcasting the ssh service.^{[2][3]}

ID: T1046

Sub-techniques: No sub-techniques

Tactic: Discovery

Platforms: Containers, IaaS, Linux, Network Devices, Windows, macOS

Contributors: Aaron Sullivan aka ZerkerEOD, Praetorian

Version: 3.2

Created: 31 May 2017

Last Modified: 15 April 2025

Version Permalink